MODULE 12 · DOMAIN 1 - ARCHITECTURE BUILT FOR THE CLOUD

Storage, Database and Caching Selection

Pick the store from the access pattern, scale it for availability, and cache what is read too often.

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Choose the store from **how the data is accessed**, not from what the data looks like.

General-purpose and analytical databases

SERVICE	ACCESS PATTERN IT FITS
Amazon RDS	Relational, an engine you already run, moderate scale
Amazon Aurora	Relational, high availability, many replicas, global reads
Amazon DynamoDB	Known keys, any scale, single-digit millisecond
Amazon DocumentDB	JSON documents, MongoDB-compatible APIs
Amazon Neptune	Relationships: fraud rings, networks, recommendations
Amazon Redshift	Analytics across large historical datasets

ON THE EXAM

Ad hoc joins mean relational. Known access patterns at massive scale mean DynamoDB.

Purpose-built stores in scope

SERVICE	CHOOSE IT WHEN	WATCH FOR
Amazon Keyspaces	Apache Cassandra workloads	CQL-compatible, serverless
Amazon Timestream	Time-series metrics and telemetry	LiveAnalytics closed to new customers
Amazon MemoryDB	In-memory speed as the primary, durable store	Not just a cache
Amazon ElastiCache	Caching in front of another store	Valkey, Redis OSS or Memcached

ON THE EXAM

Timestream for LiveAnalytics closed to new customers in June 2025. New work goes to Timestream for InfluxDB.

RDS or Aurora?

CHOOSE RDS WHEN

Familiar and economical

- Oracle, SQL Server, Db2 or specific versions
- Moderate read scaling
- Lowest cost for steady, small workloads
- Least change from on premises

CHOOSE AURORA WHEN

Availability and scale

- MySQL or PostgreSQL compatible
- Up to 15 replicas on shared storage
- Fast failover, storage grows itself
- Global Database and serverless

THE TIEBREAKER

Many readers, fast failover or cross-Region reads point to Aurora. A commercial engine points to RDS.

Scaling databases for availability

MECHANISM	WHAT IT GIVES YOU
RDS Multi-AZ instance	Synchronous standby for failover; not readable
RDS Multi-AZ DB cluster	Two readable standbys, faster failover
Read replicas	Asynchronous copies for reads, also cross-Region
Aurora Replicas	Up to 15 readers on shared storage, failover targets
Aurora Global Database	Cross-Region replicas, managed Regional failover
DynamoDB global tables	Active-active in many Regions

ON THE EXAM

Multi-AZ is for availability, replicas are for reads. A classic Multi-AZ standby serves no traffic.

Global tables: eventual or strong consistency?

MULTI-REGION EVENTUAL (MREC)

The default

- Replicas in any number of Regions
- Replication typically under a second
- Last writer wins on conflicts
- RPO of a few seconds

MULTI-REGION STRONG (MRSC)

RPO of zero

- Exactly three Regions, or two plus a witness
- Reads always return the latest write
- Higher write latency
- No TTL, LSIs or transactions

THE TIEBREAKER

RPO of zero and read-your-latest-write in any Region means MRSC. The mode cannot be changed later.

Storage is chosen by **access protocol** first. Performance and cost come second.

Storage by access pattern

SERVICE	PROTOCOL	CHOOSE IT WHEN
Amazon S3	Objects, HTTP	Durable, unlimited, most data
S3 Express One Zone	Objects, one AZ	Single-digit ms, very high request rates
EBS gp3	Block, one instance	Most volumes; up to 80,000 IOPS
EBS io2 Block Express	Block, Multi-Attach	Sub-ms, beyond 80,000 IOPS
Amazon EFS	NFS, Linux	Shared files across instances and AZs
Amazon FSx	SMB, Lustre, ONTAP, OpenZFS	Windows, HPC or multiprotocol

ON THE EXAM

Since September 2025 gp3 reaches 80,000 IOPS. The old rule, above 16,000 needs io2, is wrong.

EBS gp3 or io2 Block Express?

CHOOSE GP3 WHEN

Cost-effective performance

- Up to 80,000 IOPS and 2,000 MiB/s
- IOPS and throughput set independently
- Millisecond latency is acceptable
- Most databases and applications

CHOOSE IO2 BLOCK EXPRESS WHEN

The highest tier

- Up to 256,000 IOPS
- Sub-millisecond latency
- Multi-Attach to several instances
- 99.999 percent durability

THE TIEBREAKER

IOPS alone no longer forces io2 below 80,000. Latency, Multi-Attach or extreme IOPS still do.

Caching strategies

CACHE	SITS IN FRONT OF	CHOOSE IT WHEN
DynamoDB Accelerator	DynamoDB only	Microsecond reads, no app logic change
ElastiCache	Any database or API	Flexible caching, sessions, leaderboards
CloudFront	Web origins	Content close to users
API Gateway cache	REST API stages	Repeated identical API responses

ON THE EXAM

Lazy loading caches what is read. Write-through keeps the cache fresh. TTL bounds staleness in both.

DAX or ElastiCache?

CHOOSE DAX WHEN

DynamoDB, transparently

- The database is DynamoDB
- Microsecond read latency
- Swap the client, keep the calls
- Item and query results cached

CHOOSE ELASTICACHE WHEN

Anything, with control

- Relational or other sources
- Cache computed results or sessions
- Valkey is the lower-cost engine
- Application controls what is cached

THE TIEBREAKER

DynamoDB with minimal code change means DAX. Any other source, or computed results, means ElastiCache.

Reading the scenario

WHEN THE QUESTION SAYS	IT IS TESTING
Many readers, failover in seconds, MySQL	Aurora with Aurora Replicas
RPO zero, latest write from any Region	DynamoDB global tables, MRSC
New time-series workload	Timestream for InfluxDB
40,000 IOPS, cost-sensitive	gp3, not io2
Microsecond reads on DynamoDB	DAX
Windows SMB and Linux NFS, same data	FSx for NetApp ONTAP

ON THE EXAM

Three of these six answers changed since 2025. Check every number against today's limits.

> Meridian Freight's tracking database is MySQL on Amazon RDS. Read traffic will grow tenfold, needing around ten read replicas, and failover must complete within about 30 seconds.
>
> The team also wants storage to grow without intervention.

Which solution should the solutions architect recommend?

A RDS for MySQL Multi-AZ with ten read replicas

B Amazon Aurora MySQL with Aurora Replicas

C Migrate the data to Amazon DynamoDB

D An RDS Multi-AZ DB cluster with two readable standbys

◆ PAUSE AND DECIDE

Which solution should the solutions architect recommend?

A. RDS with ten read replicas

Replicas lag asynchronously, each with its own storage to manage, and are not automatic failover targets.

B. Aurora MySQL with Aurora Replicas

Readers on shared storage, fast failover, storage that grows itself.

C. DynamoDB

A rewrite of a relational application for a read-scaling problem.

D. Multi-AZ DB cluster

Two readable standbys cannot serve tenfold read growth.

THE SIGNAL

Many readers, fast failover and self-growing storage on MySQL. Each is an Aurora architecture property.

> Meridian Freight is building a multi-Region payments ledger. Losing any acknowledged write in a Regional failure is unacceptable, and a read in any Region must return the latest balance.
>
> The team accepts higher write latency, and the table has no TTL or transactions.

Which design meets these requirements?

A DynamoDB global tables with multi-Region eventual consistency

B Aurora Global Database with write forwarding

C DynamoDB global tables with multi-Region strong consistency across three Regions

D A single-Region DynamoDB table with point-in-time recovery

◆ PAUSE AND DECIDE

Which design meets these requirements?

A Eventual consistency

RPO of a few seconds, and reads can be stale in another Region.

B Aurora Global Database

Asynchronous replication to secondaries, so RPO is not zero.

C Strong consistency, three Regions

RPO of zero, and every Region reads the latest write.

D Single Region with PITR

Restores from backup; nothing serves other Regions.

THE SIGNAL

Zero data loss plus latest-read everywhere is MRSC. The scenario even removes its two restrictions.

> Meridian Freight is starting a new project to store engine telemetry from 20,000 trucks as time-series data, queried by time range for dashboards and alerts.
>
> Meridian has never used Amazon Timestream, and wants a managed time-series database.

Which service should the solutions architect choose?

A Amazon Timestream for InfluxDB

B Amazon Timestream for LiveAnalytics

C Amazon DynamoDB with a time-based sort key

D Amazon Redshift

◆ PAUSE AND DECIDE

Which service should the solutions architect choose?

A Timestream for InfluxDB

A managed time-series database AWS recommends for new customers.

B Timestream for LiveAnalytics

Closed to new customers since 20 June 2025.

C DynamoDB with time sort key

Possible, but you build time-series querying and retention yourself.

D Redshift

Analytics warehouse, not a store for live dashboards and alerts.

THE SIGNAL

Never used Timestream is the planted detail. New customers cannot start on LiveAnalytics.

A Meridian Freight database server needs a 4 TiB volume with a sustained 40,000 IOPS. Millisecond latency is acceptable, and the volume is attached to one instance.

The company wants the lowest-cost volume that meets the requirement.

Which EBS volume type should the solutions architect choose?

A. io2 Block Express with 40,000 provisioned IOPS

B. Throughput Optimized HDD (st1)

C. Several gp2 volumes striped with RAID 0

D. gp3 with 40,000 provisioned IOPS

◆ PAUSE AND DECIDE

Which EBS volume type should the solutions architect choose?

A io2 Block Express

Meets it, at a higher price, for latency and features nobody asked for.

B st1

Hard disk for sequential throughput, nowhere near 40,000 IOPS.

C Striped gp2

The old workaround, with more volumes to manage and IOPS tied to size.

D gp3, 40,000 IOPS

Within gp3's 80,000 IOPS ceiling since 2025, at lower cost.

THE SIGNAL

40,000 IOPS used to force io2. Since September 2025 it fits comfortably in gp3.

> Meridian Freight's rate catalogue lives in DynamoDB. During peak hours the same items are read thousands of times per second, and the application needs microsecond read latency.
>
> The development team wants to avoid writing caching logic.

What should the solutions architect add?

A. An ElastiCache for Valkey cluster with lazy loading in the application

B. A DynamoDB Accelerator (DAX) cluster, with the application using the DAX client

C. More provisioned read capacity on the table

D. DynamoDB global tables in a second Region

◆ PAUSE AND DECIDE

What should the solutions architect add?

A ElastiCache with lazy loading

Works, but lazy loading is exactly the caching logic the team wants to avoid.

B DAX

API-compatible cache for DynamoDB; microsecond reads with a client swap.

C More read capacity

More throughput, still millisecond latency.

D Global tables

For multi-Region availability, not microsecond reads.

THE SIGNAL

DynamoDB, microseconds, no caching code. All three point at DAX.

> Meridian Freight's billing application runs on Windows and needs SMB file shares joined to Active Directory. A Linux analytics cluster must read the same files over NFS.
>
> The storage must be fully managed and highly available.

Which storage should the solutions architect choose?

A Amazon FSx for Windows File Server

B Amazon EFS

C Amazon FSx for NetApp ONTAP

D S3 File Gateway

◆ PAUSE AND DECIDE

Which storage should the solutions architect choose?

A FSx for Windows File Server

SMB with Active Directory, but not NFS for the Linux cluster.

B EFS

NFS for Linux, with no SMB for Windows.

C FSx for NetApp ONTAP

Serves the same data over SMB and NFS, joined to Active Directory.

D S3 File Gateway

A hybrid gateway appliance, not managed shared storage in AWS.

THE SIGNAL

Two protocols on the same data is the defining requirement. Multiprotocol access is ONTAP's strength.

What to carry out of this module

- **Access pattern first.** Joins mean relational; known keys at scale mean DynamoDB.
- **Multi-AZ for failover, replicas for reads.** Aurora for many readers and fast failover.
- **MRSC** for RPO zero across Regions: three Regions, no TTL, LSIs or transactions.
- **gp3 up to 80,000 IOPS.** io2 for sub-millisecond, Multi-Attach or more.
- **Timestream for LiveAnalytics** is closed to new customers. Use InfluxDB.

> ON THE EXAM
>
> DAX for DynamoDB without code; ElastiCache for anything else. ONTAP when SMB and NFS share data.